1️⃣ System Architecture

Frontend

- React.js
- TailwindCSS
- React Router
- Axios
- Context API / Redux
- Socket.io client (real-time notifications)

Backend

- Node.js
- Express.js
- MongoDB Atlas
- Mongoose
- JWT Authentication
- Socket.io (real-time system)

External Services

Images
Cloudinary

Payments
Razorpay

Emails
Nodemailer

Deployment
Frontend → Netlify
Backend → Render
Database → MongoDB Atlas

2️⃣ User Roles & Permissions

Guest

- View restaurants
- Search restaurants
- View reviews

User

- Create account
- Make reservations
- Pay reservation fee
- Leave reviews
- Cancel reservation
- Receive notifications

Restaurant Owner

- Add restaurant
- Manage restaurant
- View reservations
- Reply to reviews
- Manage tables
- Requires **subscription**

Admin

- Approve restaurants
- Manage users
- Remove reviews
- View analytics
- Manage subscription plans

3️⃣ Complete Navigation Flow

```
Home
 ├ Search Restaurants
 │
 └ Restaurant Details
       ├ Menu
       ├ Reviews
       └ Book Reservation
Auth
 ├ Login
 └ Register
User Dashboard
 ├ My Reservations
 ├ My Reviews
 ├ Notifications
 └ Profile
Owner Dashboard
 ├ Subscription Plan
 ├ Add Restaurant
```

```
  ├ Manage Restaurant
  ├ Tables
  ├ Reservations
  ├ Reviews
  └ Analytics
Admin Dashboard
  ├ Restaurants
  ├ Users
  ├ Reservations
  ├ Reviews
  ├ Subscriptions
  └ Platform Analytics
```

4️⃣ Page-by-Page Feature Plan

🏠 Home Page

Purpose

Discover restaurants.

Sections

Navbar
Hero search
Popular restaurants
Top rated restaurants
Cuisine categories
Nearby restaurants
Footer

Features

Search restaurants
Filter by cuisine
Filter by location
Recommended restaurants

🔎 Search Restaurants Page

Filters

Cuisine
Price range
Location
Rating
Features

Features include

Outdoor seating
Live music
Vegetarian
WiFi
Parking

Layout

Left side → filters
Right side → restaurant cards

Restaurant Card

Image
Restaurant name
Cuisine
Rating
Price range
Location
Reserve button

🍽️ Restaurant Details Page

Sections

Restaurant banner
Restaurant info
Menu
Photo gallery
Reservation section
Reviews

Reservation Section

Inputs

Date
Time slot
Party size

System checks

Table availability

Reviews Section

Shows

User rating
Comment
Images
Owner reply

Users can

Write review
Edit review
Delete review

Owner can

Reply to review

📅 Reservation System

Reservation flow

```
User selects date
User selects time slot
User selects party size
System checks available tables
User pays reservation fee
Reservation confirmed
Notification sent
```

👤 User Dashboard

Navigation

```
My Reservations
My Reviews
Notifications
Profile
```

My Reservations

Shows

Restaurant name
Date
Time
Party size
Reservation status

Options

Cancel reservation
Modify reservation

Notifications

Shows

Reservation confirmation
Reservation reminder
Reservation cancelled
Owner message

🏪 Restaurant Owner Dashboard

Before accessing dashboard

Owner must **purchase subscription plan**

Subscription Page

Plans

Basic
Premium
Enterprise

Example

```
Basic → 1 restaurant
Premium → 3 restaurants
Enterprise → unlimited
```

Owner must pay subscription fee.

Add Restaurant

Owner adds

Restaurant name
Description
Cuisine
Price range
Location
Opening hours
Menu
Images
Features

Manage Restaurant

Owner can

Edit details
Upload images
Update menu
Add tables
Change opening hours

Tables Management

Owner defines

```
Table 1 → capacity 2
Table 2 → capacity 4
Table 3 → capacity 6
```

Reservations Page

Owner sees

Upcoming reservations
Customer name
Party size
Status

Options

Confirm reservation
Cancel reservation

Reviews Management

Owner can

Reply to reviews.

Owner Analytics

Shows

Total reservations
Revenue
Popular dishes
Peak booking times

👑 Admin Dashboard

Navigation

```
Restaurants
```

```
Users
Reservations
Reviews
Subscriptions
Analytics
```

Restaurant Management

Admin can

Approve restaurant
Reject restaurant
Delete restaurant

Users

Admin can

View users
Ban user
Delete user

Reservations

Admin can

View reservations
Cancel reservations

Reviews

Admin can

Delete fake reviews

Subscription Management

Admin manages

Subscription plans
Pricing
Owner subscriptions

Analytics

Shows

Total users
Total restaurants
Total reservations
Revenue

5️⃣ Real-Time Notification System

Use **Socket.io**

Events

```
reservation_created
reservation_cancelled
reservation_confirmed
review_added
payment_success
subscription_expiring
```

Notifications sent to

User
Restaurant Owner
Admin

Example

User books table → Owner gets notification.

6️⃣ Table Availability Logic

Restaurant has tables.

Example

```
Table 1 → capacity 2
Table 2 → capacity 4
Table 3 → capacity 6
```

Reservation request

```
party size = 3
```

System selects

```
table capacity >= party size
```

Table becomes locked for that time slot.

7️⃣ Payment System

Reservation Payment

User pays reservation fee.

Example

```
₹200 booking fee
```

Flow

```
Create order
User pays
Verify payment
Confirm reservation
```

Payment gateway

Razorpay

8️⃣ Owner Subscription System

Owners must pay monthly subscription.

Example

```
Basic → ₹999/month
Premium → ₹2999/month
```

Flow

```
Owner selects plan
Payment completed
Dashboard unlocked
```

9️⃣ Refund Policy Logic

Refund allowed if reservation cancelled before cutoff time.

Example

```
Reservation time = 8PM
Cancellation before = 6PM
```

Refund rules

```
Cancel > 2 hours before → full refund
Cancel < 2 hours before → no refund
```

Refund process

```
Cancel reservation
Check policy
Trigger Razorpay refund
```

🔟 API Structure

Authentication

```
POST /auth/register
POST /auth/login
GET /auth/me
```

Restaurants

```
POST /restaurants
GET /restaurants
GET /restaurants/:id
```

```
PUT /restaurants/:id
DELETE /restaurants/:id
```

Reservations

```
POST /reservations
GET /reservations/my
PUT /reservations/:id
DELETE /reservations/:id
```

Reviews

```
POST /reviews
PUT /reviews/:id
DELETE /reviews/:id
GET /restaurants/:id/reviews
```

Availability

```
GET /restaurants/:id/availability
```

Payments

```
POST /payments/create-order
POST /payments/verify
POST /payments/refund
```

Subscriptions

```
POST /subscriptions/create
GET /subscriptions/my
```

1️⃣1️⃣ Database Schema

Collections

```
Users
Restaurants
Tables
Reservations
Reviews
Payments
Subscriptions
Notifications
```

1️⃣2️⃣ Notifications Collection

```
Notification
{
 userId
 type
 message
 isRead
 createdAt
}
```

1️⃣3️⃣ Background Jobs

Automated tasks

Reservation reminders
Subscription expiry alerts
Daily analytics reports

Use

`node-cron`

1️⃣4️⃣ Deployment Plan

Frontend

Deploy to Netlify

Backend

Deploy to Render

Database

MongoDB Atlas

⭐ Final Project Structure

```
restaurant-reservation-platform
client
 ├ components
 ├ pages
 ├ context
 ├ services
 ├ hooks
 └ utils
server
 ├ controllers
 ├ models
 ├ routes
 ├ middleware
 ├ config
```